Spend Steps, Not Subsets: Time-Targeted Optimization for Online Gaussian Splatting Mapping

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Abstract

Online RGB-D Gaussian Splatting mappers commonly optimize under a modeled per-Gaussian budget that decides which Gaussians to update each frame. We show that with a fast GPU rasterizer the wall-clock cost of a frame is dominated by a per-step overhead of about 5.2 ms that does not depend on how many Gaussians a step updates (a fixed 16 ms per frame remains), so selecting subsets saves only about 3 ms and costs several dB. We profile the pipeline and remove its main overheads (2.0–2.4× faster at unchanged output), update every Gaussian at every step, and control the number of steps *with a feedback loop on measured frame time. The step count is only useful with an optimizer whose statistics do not go stale between frames; short-memory Adam, higher learning rates and an in-frame decay account for* $+6.98$ *dB at equal time. On 11 test scenes and five later windows of TUM sequences, the controller meets 45–90 ms targets within 3.7% and, at the original system's frame time, raises PSNR on mapped frames by 10.53 dB (worst scene* $+7.41$ *dB). On frames held out from mapping the gain is smaller,* $+1.9$ *to* $+3.3$ *dB at* $K=3$ *steps, and it saturates quickly. Value-gated growth, early stopping, error-proportional step allocation and keyframe replay do not beat uniform steps at equal time. With the tracker of MonoGS run against our map on three TUM sequences, held-out PSNR is* $+3.0$ *to* $+10.9$ *dB above original MonoGS at 1.1–1.4× speed, and trajectory error changes by* -0.27 *to* $+0.46$ *cm. Main experiments use ground-truth poses on a laptop GPU.*

1. Introduction

Online RGB-D 3D Gaussian Splatting (3DGS) [6] underlies mapping systems such as SplaTAM [5], MonoGS [10], Photo-SLAM [4] and RTG-SLAM [12]. On robots and headsets the optimization available per streaming frame is limited, so a common design budgets it: a cost model prices the update of each Gaussian and a scheduler picks the subset that fits [11, 16]. Our earlier study of such a system, which we call the *original system*, found that the modeled budget was the bottleneck.

Here we measure wall-clock time instead of the model. On a GPU rasterizer a frame costs a fixed 16 ms plus 5.2 ms for every optimization step (median over 16 scenes and windows), almost independently of how many Gaussians a step updates. Selecting a subset therefore saves little: updating all Gaussians instead of the original system's subset costs only 3 ms more at a matched step count, and gains several dB. The right unit of compute is the *step*, and the right controller adjusts the number of steps to the measured frame time.

We build on this in three parts. (1) We profile the pipeline and remove its main overheads, making it 2.0–2.4× faster at unchanged output (Sec. 4). (2) We update every Gaussian at every step and control the step count with a feedback loop on measured frame time (Sec. 5). (3) We show that this only pays off with an optimizer whose statistics do not go stale across frames: default Adam carries about 125 frames of second-moment memory, which throttles the steps as the rendering error falls; short-memory Adam with higher learning rates and an in-frame decay is worth $+6.98$ dB at equal time.

At about the original system's frame time (44 ms), the time-targeted system raises PSNR on mapped frames by $+10.53$ dB on average over 11 test scenes (worst scene $+7.41$ dB) and by $+9.90$ dB on five later windows. The controller keeps the measured time within 3.7% of 45–90 ms targets and matches the best uniform step count at equal time (± 0.2 dB), whereas one fixed step count varies from 53 to 77 ms across scenes. These are large numbers, and a reader should ask whether the original system was simply under-tuned. It partly was: with the new optimizer but its old scheduler it gains only $+3.44$ dB, so most of the gain needs both the optimizer and the removal of subset selection (Tab. 2). On frames held out from mapping the gain is much smaller, $+1.9$ to $+3.3$ dB at three steps, and it saturates (Tab. 4); extra steps mostly fit the current view.

Contributions. (i) A wall-clock cost analysis showing that per-step overhead, not the number of updated Gaussians, determines frame time, with core changes that make

the pipeline about 2× faster at identical output. (ii) A time-targeted step controller and an optimizer setting that make the added steps useful, with all constants set on four development scenes. (iii) A matched-time evaluation on 11 scenes and five fresh windows against a uniform-step curve, an optimizer ablation, a held-out-view check, and negative results for value-gated growth, early stopping, error-proportional allocation and keyframe replay.

2. Related Work

Online 3DGS SLAM and dense mapping. GS-SLAM [19] and SplaTAM [5] introduced differentiable 3D-Gaussian SLAM, with SplaTAM using silhouette-guided optimization; Gaussian-SLAM [20] adds a sub-map design and Loop-Splat [22] adds loop closure by registering Gaussian sub-maps; Point-SLAM [13] uses dynamic neural point clouds; MonoGS [10] and Photo-SLAM [4] push toward real-time operation with loop closure; RTG-SLAM [12] adds compact opaque/transparent roles with rule-based stable/unstable selection and is our primary external policy baseline (re-implemented under exactly matched budgets, since native-system comparison would confound renderer, tracker, and scheduler effects). Unlike these systems, our object of study is not the full SLAM stack but the compute-allocation layer inside it; we evaluate mapping under ground-truth poses with tracking explicitly decoupled (Sec. 7). **Scalable and efficient Gaussian mapping.** Recent systems address online map scalability through representation and update design. VTGaussian-SLAM [2] ties Gaussians to RGB-D views and avoids keeping every primitive learnable, targeting large scenes and memory use. SEGS-SLAM [18] improves structure and appearance consistency, while DyGS-SLAM [3] focuses on dynamic-scene localization and reconstruction. In parallel, SteepGS [17] derives compact density-control updates for 3DGS reconstruction. A recent online-mapping preprint studies geometry-aware initialization and densification under short optimization schedules [8]. These methods make clear that map representation, density control, and online optimization are coupled. Our study asks how growth rate and update policy should respond to coverage and observable scene statistics under a fixed per-frame budget and Gaussian representation. We do not claim density control or efficient Gaussian mapping as new in general. The per-scene SOTA numbers in Appendix Supp. Tab. S2 are protocol-mismatched context. **Density control for 3DGS.** Taming 3DGS [9] uses guided, constructive densification under resource limits; Mini-Splatting [1] targets a constrained Gaussian count; Pixel-GS [21] reweights the densification gradient by covered pixels; and 3DGS-MCMC [7] replaces heuristic densification and pruning with relocation. These methods address offline reconstruction. Our setting differs in that a per-frame optimization budget is shared between existing and newly spawned Gaussians. We do not run them (Sec. 7), but compare against a size-matched Gaussian-count budget in the supplement. **Evaluation protocols in dense SLAM.** Published systems report full-sequence train-view rendering with global refinement at full resolution (Replica 33–42 dB for MonoGS/Point-SLAM/RTG-SLAM; TUM fr2_xyz 16–25 dB for MonoGS/SplaTAM/Photo-SLAM) plus full-trajectory ATE. Our protocol is deliberately narrower — budgeted online mapping on fixed horizons — so cross-paper numbers are contextual range, never superiority claims (Supp. Tab. S2). **Budgeted selection.** Our earlier work and the original system studied here choose which Gaussians to update under a modeled per-Gaussian budget [11, 16]. We show that, once the renderer is fast, this selection saves little wall-clock time.

3. Setting and Protocol

Mapper. Per frame the mapper renders the current map, forms per-pixel color and depth residuals, spawns Gaussians where the residual is high (at most 500 per frame), runs K optimization steps (Adam on color L1, SSIM and depth L1, sparse-aware state per Gaussian), and prunes. The *original system* runs at most $K{=}5$ steps and chooses, by a fractional knapsack on raw rendering error, which Gaussians receive how many of them so that a modeled cost stays below $B{=}15$ ms. The rasterizer is `gsplat`. Poses are ground truth; a closed-loop tracker check is in the supplement.

Timing. Frame time is the CUDA-synchronized latency of one `process_frame` call on a laptop GPU, including per-frame evaluation (about 4 ms), at 320×240. It excludes tracking.

Data and protocol. TUM RGB-D [15] and Replica [14]; two seeds. Constants were set on four *development* scenes (`fr1_desk`, `fr3_structure_texture_near`, `replica_room1`, `replica_office2`). The *test* set is 11 other scenes (150 frames for `fr2_xyz`, `fr1_xyz`; 99 for the rest, as the local Replica copies have 100–120 frames), and five *fresh* windows of 150 later frames of long TUM sequences. The observation that steps matter first appeared while profiling `fr2_xyz`, a test scene; the development scenes and fresh windows limit this exposure but do not remove it. For *held-out views*, every eighth frame is rendered but never mapped.

Equal-time comparison. Methods differ in frame time, so we compare each method with the *uniform-K curve*: the original mapper with all Gaussians active and $K \in \{3, 6, 10, 15\}$ steps per frame and the same optimizer, linearly interpolated in measured time at the method's own mean time.

4. Where the Time Goes

Fitting frame time as $a + bK$ over $K \in \{3, 6, 10, 15\}$ gives a median $a{=}16$ ms and $b{=}5.2$ ms per step (range 4.1–5.9);

Scene	ms before	ms after	speed-up	PSNR before	PSNR after
fr2_xyz	86	41	2.1×	26.00	26.08
fr1_xyz	105	44	2.4×	22.02	22.01
office0	90	45	2.0×	30.61	30.59

Table 1. **Core speed-ups at unchanged output.** Original budgeted configuration on the code revision before and after the changes of Sec. 4; end-to-end frame time in ms (including per-frame evaluation), mean PSNR (dB), two seeds, 100 frames.

at the original $K{=}5$, 39% of the time is the fixed part. GPU profiling at $K{=}8$ showed three avoidable costs, which we removed without changing the computation:

SSIM windows. The 11×11 Gaussian window was applied as a dense depthwise convolution whose backward pass took a third of GPU time; the window is separable, so we apply it as two 1-D passes (value and gradient agree to 10^{-6}). *Covariance.* $\Sigma = RSS^\top R^\top$ via batched 3×3 GEMMs cost about 20 ms per frame in cuBLAS; the same product as broadcast elementwise operations is several times cheaper in forward and backward. *Indexing.* With every Gaussian active, the gather of parameters, the scatter of gradients and the optimizer's gather/scatter of moments are pure overhead; a dense path skips them (equal to the sparse path to 10^{-6} in tests). A per-Gaussian device synchronization in the state store was vectorized.

Together these make the original configuration 2.0–2.4× faster at the same PSNR (Tab. 1); they apply to every method below.

5. Time-Targeted Steps

Uniform steps. All Gaussians are active at every step, and the modeled budget is held at a large constant so that it never binds. The number of steps becomes the compute knob.

Time targeting. With frame time τ_t and target T, the step count n follows $n \leftarrow \mathrm{clip}\big(n\,(T/\hat{\tau})^{\gamma},\, n_{\min},\, n_{\max}\big)$, where $\hat{\tau}$ is an exponential average of τ_t (weight 0.5), $\gamma{=}0.5$, $n \in [2, 24]$ and the integer used is $\mathrm{round}(n)$. A compact map or a fast scene automatically receives more steps, a large map fewer.

Short-memory optimizer. Adam's second-moment estimate with $\beta_2{=}0.999$ remembers about 125 frames at 8 steps per frame, while the rendering error shrinks as the map fits; stale large estimates then keep steps small. We use $(\beta_1, \beta_2) = (0.5, 0.8)$, learning rates ×6 (scale ×12) of the original ones, and a linear decay within each frame from 1.4× to 0.6×. These came from coordinate search on the four development scenes (Sec. 6); they are one setting for all scenes and both datasets. Value-gated growth, an additional controller from our earlier version that throttles spawning when recently spawned Gaussians are not worth keeping, is described in the supplement and is optional here.

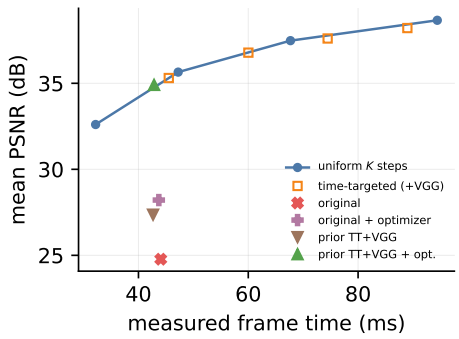


Figure 1. Mean PSNR versus measured frame time on the 11 test scenes. The original system lies far below the uniform-K curve; the time-targeted controller sits on the curve at the requested time.

6. Experiments

Against the original system (Tab. 2, Fig. 1). At matched frame time the time-targeted system with $T{=}45$ ms ends at 46 ms against 44 ms and is +10.53 dB higher on average, +7.41 dB on the worst scene, and +9.90 dB on the fresh windows. Even $K{=}3$ steps, at 32 ms, are +7.83 dB. At matched time the original system is −10.27 dB relative to the uniform-K curve. Applying the new optimizer to the original scheduler gains +3.44 dB, still far below the curve; applying it to the earlier time-targeted budget controllers (supplement) reaches the curve (+0.13 dB), because there the budget grows until every Gaussian is updated. The subset scheduler is therefore not what limits quality once the budget is large; it is what keeps the original system off the curve.

Time targeting. For $T \in \{45, 60, 75, 90\}$ ms the mean measured time is within 0.9% of the target (at most 3.7%) across 16 scenes and windows. A fixed $K{=}10$ varies from 53 to 77 ms across the same units, while the controller at $T{=}75$ stays within 73–75 ms. At equal time it is neither better nor worse than the uniform-K curve (0.08 dB at $T{=}60$ without value-gated growth). Its value is meeting a latency target without per-scene calibration, not extra quality.

Optimizer ablation (Tab. 3). Removing the short-memory betas costs 4.03 dB at $K{=}10$, the learning rates 4.53 dB and the in-frame decay 1.89 dB; removing all three costs 6.98 dB. The step count only pays with the right optimizer.

Held-out views (Tab. 4). All three configurations improve held-out PSNR on every scene, but by much less than mapped-frame PSNR: $K{=}3$ keeps 36–41% of its mapped gain and $K{=}10$ only 24–29%. Moving from $K{=}3$ to $K{=}10$ adds about 0.5 dB held out but 4–5 dB on mapped frames. Held-out frames lie on the mapped trajectory, so this does not test new trajectories. We therefore regard the held-out column, not the mapped-frame column, as the estimate of

	11 test scenes				5 fresh windows		
Method	ms	ΔPSNR mean	median	worst	ms	ΔPSNR mean	vs. uniform K at equal time (test)
Original budgeted system	44	0.00	0.00	0.00	42	0.00	−10.27
+ steps-regime optimizer	44	+3.44	+3.43	+1.81	42	+3.50	−6.78
Prior TT+VGG controllers	43	+2.59	+2.91	+0.01	38	+1.19	−7.36
Prior TT+VGG + optimizer	43	+10.13	+10.39	+7.42	39	+9.07	+0.13
Uniform K=3 steps	32	+7.83	+8.03	+5.54	29	+6.48	—
Uniform K=6 steps	47	+10.88	+10.92	+8.07	43	+9.92	—
Uniform K=10 steps	68	+12.70	+12.94	+9.86	62	+11.75	—
Uniform K=15 steps	94	+13.89	+14.16	+11.17	85	+12.50	—
Time-targeted steps, T=60	60	+12.11	+12.38	+9.04	60	+11.38	+0.08
+ VGG, T=45	46	+10.53	+10.69	+7.41	45	+9.90	+0.03
+ VGG, T=60	60	+12.01	+12.33	+8.91	60	+11.29	−0.02
+ VGG, T=75	74	+12.83	+13.44	+9.90	74	+11.89	−0.17
+ VGG, T=90	89	+13.44	+13.87	+10.69	89	+12.32	−0.11

Table 2. **Steps regime versus the original budgeted system.** PSNR change in dB on the mapped frames relative to the original system (unthrottled raw-error selection, 5 steps, modeled 15 ms budget) on the same code; mean measured end-to-end frame time in ms. The last column is the mean PSNR difference to the uniform-K curve (linear interpolation in measured time) at the method's own mean time; the original system lies far below that curve. Two seeds per scene; "test" are the 11 scenes of the earlier protocol, "fresh" are five later windows of long TUM sequences. All optimizer constants were set on four development scenes.

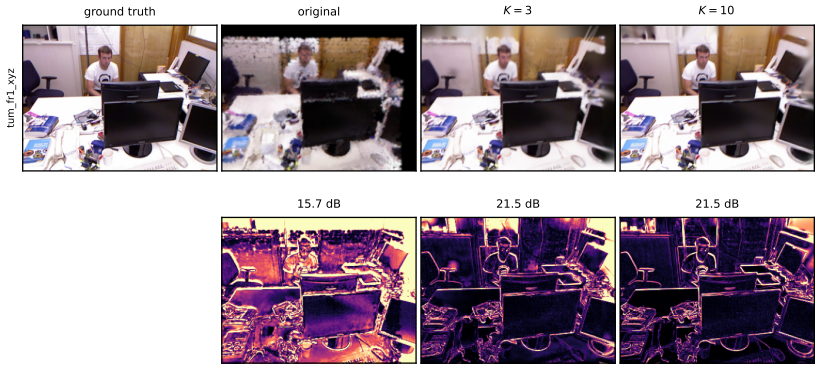


Figure 2. **Held-out view** (`fr1_xyz`, mapped frames 1–80, the next frame rendered but never mapped). Top: renders; bottom: absolute color error with PSNR. Three steps already remove most of the original system's holes and blur; ten steps add little. A second scene is in the supplement.

Optimizer	fr2	fr1	off0	off1	mean	Δ
full	37.61	35.28	41.97	43.76	39.65	0.00
default betas	32.59	30.58	38.81	40.52	35.62	−4.03
default lr	34.22	31.21	36.76	38.32	35.13	−4.53
no lr decay	35.14	32.64	40.85	42.42	37.76	−1.89
none of three	30.24	27.26	35.93	37.26	32.68	−6.98

Table 3. **Optimizer ablation** at uniform K=10 steps (about 68 ms), mean PSNR (dB) on mapped frames, two seeds, four test scenes (fr2, fr1: TUM xyz sequences; off0, off1: Replica).

		fr2	fr1	off0	off1
original	mapped / held-out	24.8 / 24.0	21.5 / 20.5	30.2 / 26.9	33.7 / 30.9
K=3	Δ mapped	+7.8	+8.1	+7.2	+5.5
	Δ held-out	+2.9	+3.3	+2.9	+1.9
K=10	Δ mapped	+12.5	+13.5	+11.3	+9.8
	Δ held-out	+3.5	+3.6	+3.3	+2.3
TT-S+VGG, T=75	Δ mapped	+12.9	+13.7	+11.5	+10.0
	Δ held-out	+3.4	+3.5	+3.2	+2.4

Table 4. **Held-out views.** Every eighth frame is rendered but never mapped. PSNR (dB) of the original system and the change for three configurations; two seeds. Gains on held-out frames are positive on every scene but a fraction of the gain on mapped frames.

map quality.

Closed-loop tracking (Tab. 5). With poses from a simple map-based tracker that aligns each frame to the current map

Scene	orig. dB	ours dB	orig. ATE	ours ATE
fr2_xyz	27.4	37.6	7.3 cm	14.0 cm
fr1_xyz	23.6	35.3	6.4 cm	5.3 cm
fr1_rpy	21.6	35.5	12.9 cm	11.6 cm
fr3_sitting_static	29.6	37.9	13.8 cm	19.4 cm
fr1_room†	14.9	24.7	154.8 cm	166.6 cm

Table 5. **Closed-loop map-based tracking in the steps regime**, three seeds, TUM, "ours" = time-targeted steps at 60 ms. PSNR is measured at the tracked pose. †Tracking fails for both (excluded from the summary).

Scene	System	PSNR	ATE	ms/frame
fr1_desk	MonoGS	18.9	1.57 cm	905
	MonoGS tracker + original mapper	17.6	1.58 cm	577
	MonoGS tracker + ours, K=3	22.7	1.37 cm	636
	MonoGS tracker + ours, K=10	23.4	1.33 cm	787
fr2_xyz	MonoGS	20.3	0.59 cm	592
	MonoGS tracker + original mapper	24.2	0.60 cm	420
	MonoGS tracker + ours, K=3	31.3	0.32 cm	529
	MonoGS tracker + ours, K=10	31.5	0.35 cm	718
fr1_xyz	MonoGS	20.3	1.07 cm	928
	MonoGS tracker + original mapper	19.9	1.06 cm	609
	MonoGS tracker + ours, K=3	23.3	1.53 cm	698
	MonoGS tracker + ours, K=10	23.5	2.10 cm	853

Table 6. **MonoGS tracking with our mapper.** Original MonoGS (RGB-D, single thread) versus MonoGS's tracker run against the map built by our mapper, first 150 frames of each sequence at 640×480, one run each. PSNR: MonoGS on non-keyframes, ours on every eighth frame, which is tracked but never mapped (both rendered at the estimated pose). ATE: Umeyama-aligned RMSE, MonoGS on its keyframes, ours on every fifth frame. Time: MonoGS total per frame; ours tracking plus mapping.

(supplement), the time-targeted system raises PSNR at the tracked pose by +8.4 to +14.0 dB on four TUM scenes, but trajectory error does not improve reliably: it changes by -1.3 to +6.7 cm and is worse on 2 of the four scenes. A map fitted tightly to a drifting trajectory is a worse alignment target, so better map fit does not imply better tracking.

MonoGS tracking with our mapper (Tab. 6). To test the mapper inside a real system, we run the pose tracker of MonoGS [10] (100 photometric plus depth iterations per frame, its own rasterizer) against the map built by our mapper, which receives the estimated pose; MonoGS's keyframe mapping, window optimization and loop handling are not used. On the first 150 frames of three TUM sequences at 640×480, held-out PSNR with K=3 steps is +3.0 to +10.9 dB above original MonoGS, and the system is 1.1–1.4× faster per frame, but the per-frame time is 88% tracking, which our mapper does not change. Trajectory error changes by -0.27 to +0.46 cm: better on two sequences, worse on fr1_xyz. With our *original* mapper instead, PSNR is within -1.3 to +3.8 dB of MonoGS, so the gain comes from the steps regime and not from replacing the mapper per se. The protocols for PSNR and ATE differ between the two systems (caption), there is one run per setting, and more steps (K=10) raise tracking error on fr1_xyz.

Relation to our earlier analysis. The earlier version of this work argued that a fixed modeled budget dilutes as the map grows and that raising it in proportion to map size improves quality on 15 scenes. Those results, and controllers that scale the modeled budget or throttle spawning, are in the supplement (section "Earlier analysis of the original budgeted system); they are consistent with Sec. 4 (the budget was starving the optimizer) but they measured a regime that the faster core and the new optimizer make obsolete.

What did not help. Evaluated on the development scenes with a single seed (details and bundles in the supplement), none of the following beat uniform steps at equal time: value-gated growth (−0.17 dB at T=75 on test, negative on fresh windows), convergence-gated early stopping (-0.15 to -0.29 dB), error-proportional step allocation across frames (-0.32 and -0.70 dB for either sign), and replaying stored keyframes in a third of the steps, which closes the mapped–held-out gap but gains only +0.23 dB held out for -3.36 dB on mapped frames. Loss-weight, spawn-rate and densification-threshold sweeps were flat.

7. Limitations

Scope. Poses are ground truth in the main results; closed-loop checks with a simple map-based tracker and with the tracker of MonoGS are TUM only (the local Replica copies are sampled too sparsely for any tracker), use one run per setting for MonoGS, and do not include MonoGS's keyframe mapping, bundle adjustment or loop closure. We use 320×240 frames, up to 150 frames per run, a laptop GPU and two seeds that measure repeatability, not scene variance; frame times of 30–90 ms are 11–30 FPS, so the high-quality settings are not real-time at 30 FPS. **Generalization.** Mapped-frame PSNR is dominated by fitting the current view; held-out gains are 24–41% of it, and held-out frames lie on the mapped trajectory. **Baselines.** The original system was strongly under-optimized; we show the ladder from it to the new system, but we do not run offline density-control methods [1, 7, 9, 21], and only the tracker of a full SLAM system is integrated. **Exposure.** The development scenes were used to set every constant, but the key observation was first made on a test scene, and constants for value-gated growth are inherited from the earlier version and were not retuned.

8. Conclusion

When the renderer is fast, the cost of a frame is set by the number of optimization steps, not by how many Gaussians a

step updates. Updating all Gaussians, controlling the number of steps from measured frame time, and using an optimizer with short memory gives $+10.53$ dB on mapped frames at the original system's frame time and $+1.9$ to $+3.3$ dB on held-out frames at lower cost, with the time target met to within 3.7%. With MonoGS's tracker running against our map, held-out PSNR is $+3.0$ to $+10.9$ dB above the original system at similar trajectory error. The remaining gap between mapped and held-out quality is the next problem.

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